Flexible Deviations from FIRE in the Sequence Space

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Motivation

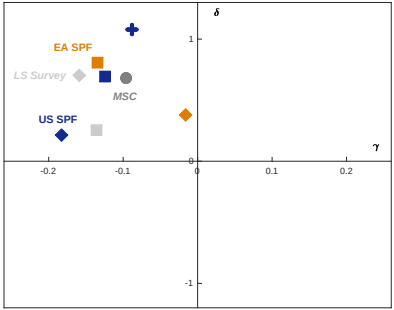
- **Question**: Can we depart from FIRE in a *flexible* and *standardised* manner that is consistent with the empirical literature (macro, micro and behavioral)?
- Recent HANK literature emphasises the importance of matching microeconomic evidence/moments
  - Auclert et al. (2020) also emphasise need to simultaneously match the macro evidence i.e. macro humps.
  - Their solution is sticky expectations which creates *persistence* but tends to kill *amplification*
  - This can be an issue e.g. Albuquerque et al. (2025).
- Growing empirical evidence base on the distance between actual expectations and rational expectations e.g. Adam et al. (2024),Coibion and Gorodnichenko (2015), Kohlhas and Walther (2021).

Idea/Proposal

- Propose that we can depart from FIRE in a similar way as in Auclert et al. (2020) but based on the reduced form used in Kohlhas and Walther (2021):

$$x_{t+k} - f_{it}x_{t+k} = \alpha_i + \underbrace{\gamma x_t}_{Today} + \underbrace{\delta(\bar{f}_t x_{t+k} - \bar{f}_{t-1} x_{t+k})}_{News} + \epsilon_{i,t|t+k}$$

Figure 1: Empirical evidence on δ and γ



Note: Evidence from Kohlhas and Walther (2021)

Idea/Proposal

What are the advantages of this approach:

1. Clear mapping from the empirical literature.
2. Allows for fact agents might simultaneously overreact to outcomes and underreact to news (Figure 1)
   - And generate dynamic overshooting property (Angeletos and Huo (2021))
3. Nests different expectations models:
   - Asymmetric attention ($\gamma, \delta \neq 0$)
   - Sticky / noisy information ($\gamma = 0, \delta > 0$)
   - $\approx$ Diagnostic expectations ($\delta = 0, \gamma < 0$)
4. Is near-rational expectations with agents behaving rationally conditional on their subjective beliefs.

Related Literature

Empirical evidence on expectation deviations:

- Information rigidity: Coibion and Gorodnichenko (2015)
- Asymmetric attention / extrapolation: Kohlhas and Walther (2021)
- Diagnostic expectations: Bordalo et al. (2018)
- Dynamic overshooting: Angeletos and Huo (2021); Adam et al. (2024)

Sequence Space:

- Sequence-space Jacobian methods: Auclert et al. (2021)
- Non-FIRE expectations in sequence space: Bardóczy (2023)
- Micro jumps, macro humps (estimated HANK): Auclert et al. (2020)

Estimating DSGE models with survey expectations:

- SPF expectations as observables improve model fit and discrimination: Del Negro and Eusepi (2011)
- Survey forecasts improve out-of-sample DSGE forecasting: Del Negro and Schorfheide (2013)
- Adaptive learning in SW model: beliefs closer to surveys, lower shock persistence: Slobodyan and Wouters (2012)

This Paper

Contribution: Implements both underreaction *and* overreaction in the sequence space — parsimoniously, with two reduced-form parameters (δ, γ).

1. **Flexible framework**: nests FIRE, sticky information, noisy information, and extrapolative expectations as special cases — agnostic on micro-foundations
2. **Sequence-space solvent**: derives forecast matrices Λ_h and behavioural Jacobian $\hat{J}$ as a drop-in replacement for FIRE Jacobians, applicable block-by-block to any DSGE or HANK model
3. **HANK estimation with survey data**: estimates a two-asset HANK model using SPF *and* Michigan Survey inflation expectations to separately identify firm and household behavioural parameters
4. **Findings**: extrapolation amplifies humps and supports dynamic overshooting; improves fit with the data; but combination of model and data favours high stickiness for households and firms, in tension with micro evidence.

Sequence Space Jacobian: Recap (Auclert et al., 2021)

Key idea: Solve DSGE models by linearising around the steady state in the space of length-T sequences, rather than recursively.

Any model can be written as $\mathbf{F}(\mathbf{U}, \mathbf{Z}) = \mathbf{0}$. Linearising gives the **GE Jacobian** $\mathbf{G}$ — each column is a full IRF:

$$d\mathbf{U} = \underbrace{-\mathbf{H}_U^{-1}\mathbf{H}_Z}_{\equiv \mathbf{G}} \, d\mathbf{Z}$$

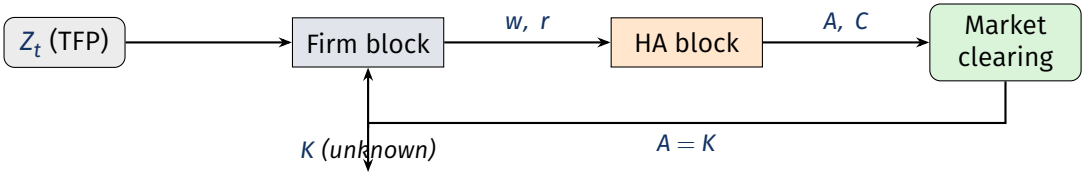
The model is decomposed into blocks linked by a directed acyclic graph (DAG). Each block maps input sequences $d\mathbf{X}$ to output sequences $d\mathbf{Y}$ via a **block Jacobian** J:

$$d\mathbf{Y} = J \, d\mathbf{X}, \qquad J_{s,t} = \frac{\partial Y_s}{\partial X_t}$$

Chaining Js across blocks along the DAG recovers $\mathbf{G}$ efficiently via the fake-news algorithm (Auclert et al., 2021).

Our extension: replace J with the behavioural Jacobian $\hat{J}$ in every forward-looking block.

Example DAG: Krusell–Smith Model



- Each arrow carries a block Jacobian J: the PE response of downstream outputs to upstream inputs
- J_{HA} maps (r, w) sequences $\longrightarrow$ savings A and consumption C sequences
- Market clearing ($A = K$) closes the loop and pins down the unknown K sequence
- **Our extension:** $J_{\text{HA}} \longrightarrow \hat{J}_{\text{HA}}$ — households form expectations r^e, w^e rather than observing r, w directly

Implementing Behavioural Expectations: Overview

Key steps:

1. Start from the **KW reduced-form** forecast error regression — two parameters (γ, δ)
2. Average across agents and rearrange to obtain a **recursive forecast law of motion** for the average forecast
3. Build **forecast matrices** Λ_h that map the realised aggregate path $d\mathbf{X}$ into time-h expectations $d\mathbf{X}^{e,h}$
4. Construct the **behavioural Jacobian** $\hat{J}$ from Λ_h and the FIRE Jacobian J
5. **Replace FIRE Jacobians block by block** in the general equilibrium solvent — apply different (θ, γ) to households and firms

Key assumptions (ensure internal rationality):

1. **No steady-state bias**: $c = \frac{\gamma}{1+\delta} x_{ss}$ — expectations are unbiased around the steady state
2. **Law of iterated expectations**: agents do not forecast their own forecast errors; future errors are orthogonal to current information sets
3. **Terminal condition**: for large enough T, agents expect the economy to return to steady state — guarantees asymptotic rationality
4. **Stability**: $\delta > 0$ (equivalently $\theta \in (0, 1)$) — ensures the forecast LoM is stable and forecast errors eventually vanish

Framework I: The Kohlhas-Walther Reduced Form

Start from the forecast error regression of Kohlhas and Walther (2021):

$$\underbrace{x_{t+k} - f_{it}x_{t+k}}_{\text{forecast error}} = \alpha_i + \underbrace{\gamma x_t}_{\text{extrapolation}} + \underbrace{\delta(\bar{f}_t x_{t+k} - \bar{f}_{t-1}x_{t+k})}_{\text{news revision}} + \varepsilon_{i,t|t+k}$$

Parameter γ: overreaction to current conditions

- $\gamma < 0$: agents extrapolate from x_t $\Rightarrow$ too pessimistic in recessions, too optimistic in booms
- $\gamma = 0$: no extrapolation

Parameter δ: underreaction to news

- $\delta > 0$: slow incorporation of news about future x
- $\delta = 0$: full attention

Nested models: FIRE ($\gamma = \delta = 0$), sticky/noisy information ($\gamma = 0$, $\delta > 0$), pure extrapolation ($\gamma < 0$, $\delta = 0$), asymmetric attention (both non-zero).

Framework II: Behavioural Forecast Law of Motion

Averaging the KW regression and rearranging yields a recursive forecast LoM:

$$\bar{f}_t dx_{t+k} = \underbrace{\frac{\delta}{1+\delta}}_{\equiv \theta} \bar{f}_{t-1} dx_{t+k} + \underbrace{\frac{1}{1+\delta}}_{1-\theta} (dx_{t+k} - \gamma\, dx_t)$$

Economic interpretation:

- Fraction θ do **not** update $\Rightarrow$ carry over last period's forecast (Calvo-style rigidity)
- Fraction $(1-\theta)$ **update** but extrapolate from current x_t with bias $-\gamma\, dx_t$

Framework II: Behavioural Forecast Law of Motion

Forecast error law of motion (for any future period $t+k$):

$$\underbrace{h_t^{t+k}}_{\text{forecast error}} = \theta \underbrace{h_{t-1}^{t+k}}_{\text{prior error}} + (1-\theta) \underbrace{\gamma\, dx_t}_{\text{extrapolation from today}}$$

Key synergy: θ sustains persistence (delayed peak); $\gamma < 0$ adds amplification (larger peak). Together they generate large hump-shaped responses.

Framework III: Forecast Matrices Λ_h

Solve the recursive forecast LoM forward from the steady state. The time-$t+h$ forecast of dx_{t+k} can be written as:

$$d\bar{f}_{t+h} dx_{t+k} = (1-\theta) \sum_{j=0}^{h} \theta^{t-j} \left(dx_{t+k} - \gamma \, dx_{t+j} \right)$$

Stack these into a **forecast matrix** $\Lambda_h \in \mathbb{R}^{T \times T}$ mapping the realised path $d\mathbf{X}$ to time-h forecasts $d\mathbf{X}^{e,h}$:

$$\Lambda_h = \begin{pmatrix} \mathbf{I}_{h+1} & \mathbf{0} \\ \mathbf{A}_h & \mathbf{B}_h \end{pmatrix}$$

Framework III: Forecast Matrices Λ_h

$$\Lambda_h = \begin{pmatrix} \mathbf{I}_{h+1} & \mathbf{0} \\ \mathbf{A}_h & \mathbf{B}_h \end{pmatrix}$$

Top-left $\mathbf{I}_{h+1}$: past and current inputs observed perfectly — no distortion of history.

$\mathbf{A}_h$ (extrapolation block):

$$A_{h(i,j)} = \frac{-\gamma\,\theta^{h-j}}{1-\theta}, \quad j = 0, \ldots, h$$

Collects the extrapolation bias each cohort of updaters incurs at the time they last updated ($t-j$), weighted by the measure θ^{h-j} still using that vintage.

$\mathbf{B}_h$ (news pass-through block):

$$B_{h(i,i)} = \sum_{l=0}^{h} (1-\theta)\,\theta^{l} \xrightarrow{h\to\infty} 1$$

Diagonal — accumulated probability of having received news about dx_k by time h. Converges to the identity as $h \to \infty$ (asymptotic rationality).

Framework III: Behavioural Jacobian & GE Solvent

With the forecast matrices in hand, construct the **behavioural Jacobian** $\hat{J}$ at the block level (Bardóczy, 2023):

$$\hat{J} = J\Lambda_0 + \sum_{h=1}^{T} R_h(\Lambda_h - \Lambda_{h-1}), \qquad R_h = \begin{pmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & J \end{pmatrix}_{\text{shifted by } h}$$

Intuition: outcomes of Y depend on a sequence of expectation revisions as time progresses. Each revision — weighted by $(\Lambda_h - \Lambda_{h-1})$ — is an unanticipated update to which the FIRE Jacobian J is re-applied at the appropriate horizon via R_h.

Application: Two-Asset HANK Model & Estimation

Model (Auclert et al., 2020):

- 6-group household heterogeneity (liquid + illiquid assets)
- Sticky prices & wages (Calvo)
- Investment with time-to-build + adjustment costs
- Long-term government debt
- Taylor rule + fiscal stabilisation

Estimation: Bayesian full-information MLE

- 9 quarterly observables (1978–2018): GDP, C, I, hours, wages, inflation, FFR, **SPF expectations** (firms), **Michigan Survey** (households)
- 7 AR(1) shock processes (+ MA terms for markup shocks)
- 4 dynamic parameters (ψ, ρ_m, ζ_p, ζ_w)
- **4 behavioural parameters**: θ_{hh}, θ_{prof}, γ_{hh}, γ_{prof}

Application: Two-Asset HANK Model & Estimation

We will estimate the model in 3 ways:

1. Estimate the model with FIRE expectations ($\theta = 0$, $\gamma = 0$) — benchmark
2. Estimate the model with parameters from the Literature — plug in θ and γ from micro/empirical studies, without using survey data — calibration
3. Estimate the model with free behavioural parameters — full estimation
   - With and without survey expectations

Parameters to Estimate

Shock Processes

Param.	Description	Prior
Persistence		
ρ_c	Discount factor	Beta(0.5, 0.2)
ρ_r	Monetary policy	Beta(0.5, 0.2)
ρ_g	Govt. spending	Beta(0.5, 0.2)
ρ_z	TFP	Beta(0.5, 0.2)
ρ_{rp}	Risk premium	Beta(0.5, 0.2)
ρ_p	Price markup	Beta(0.5, 0.2)
ρ_w	Wage markup	Beta(0.5, 0.2)
Standard deviation		
$\sigma_c, \sigma_r, \sigma_g, \sigma_z$		Inv-Gamma(0.1, 2)
$\sigma_{rp}, \sigma_p, \sigma_w$		Inv-Gamma(0.1, 2)
MA terms (markup shocks)		
θ_p	Price markup MA(1)	Beta(0.5, 0.2)
θ_w	Wage markup MA(1)	Beta(0.5, 0.2)

Dynamic Parameters

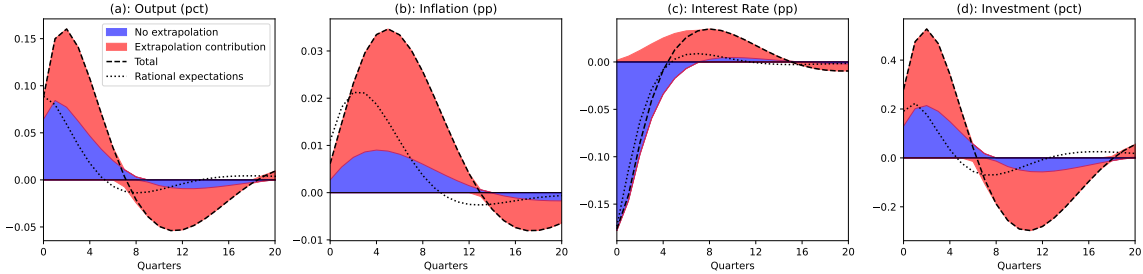
Param.	Description	Prior
ψ	Inv. adj. cost	Normal(10, 4)
ρ_m	Taylor rule inertia	Beta(0.5, 0.2)
ζ_p	Calvo price prob.	Beta(0.7, 0.2)
ζ_w	Calvo wage prob.	Beta(0.7, 0.2)

Behavioural Parameters

Param.	Description	Prior
θ_{hh}	HH stickiness	Beta(0.5, 0.2)
θ_{prof}	Firm stickiness	Beta(0.5, 0.2)
γ_{hh}	HH extrapolation	Normal(0, 0.3)
γ_{prof}	Firm extrapolation	Normal(0, 0.3)

Results: Behavioural Properties - Amplification & Persistence

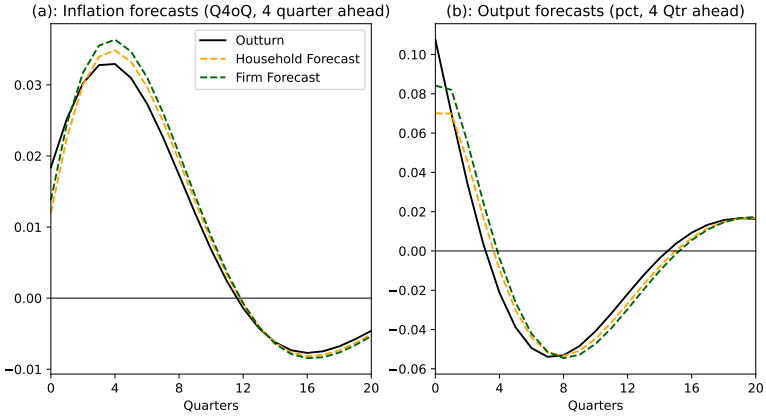
Figure Impulse response to a monetary policy shock



- Where we set $\theta_{hh} = 0.4$ and $\theta_{prof} = 0.3$,$\gamma_{hh} = -0.10$ and $\gamma_{prof} = -0.15$ based on the micro/empirical literature.

Results: Behavioural Properties - Underreaction & Overreaction

Figure Expectations and outurns (Impulse response to a monetary policy shock)



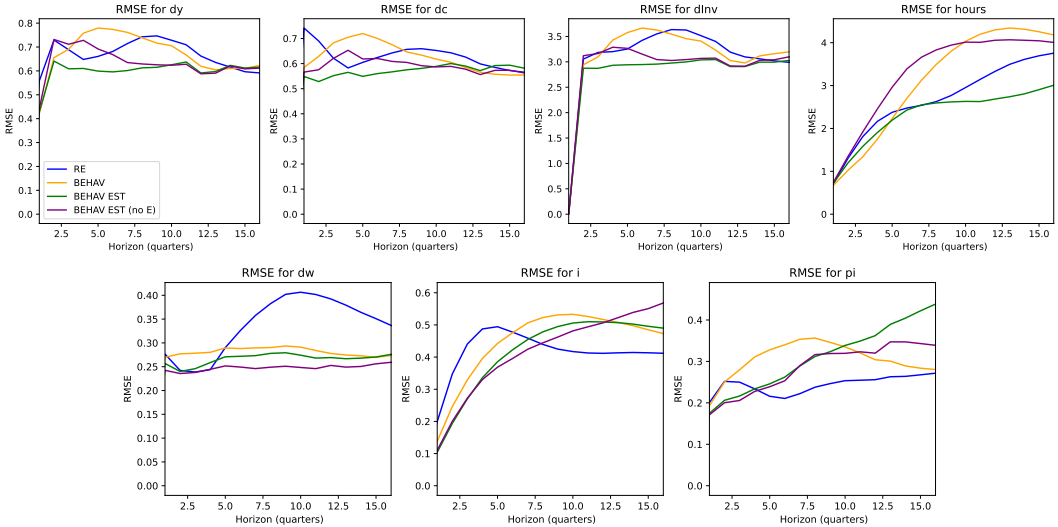
- Where we set $\theta_{hh} = 0.4$ and $\theta_{prof} = 0.3$,$\gamma_{hh} = -0.10$ and $\gamma_{prof} = -0.15$ based on the micro/empirical literature.

Results: Model Comparison

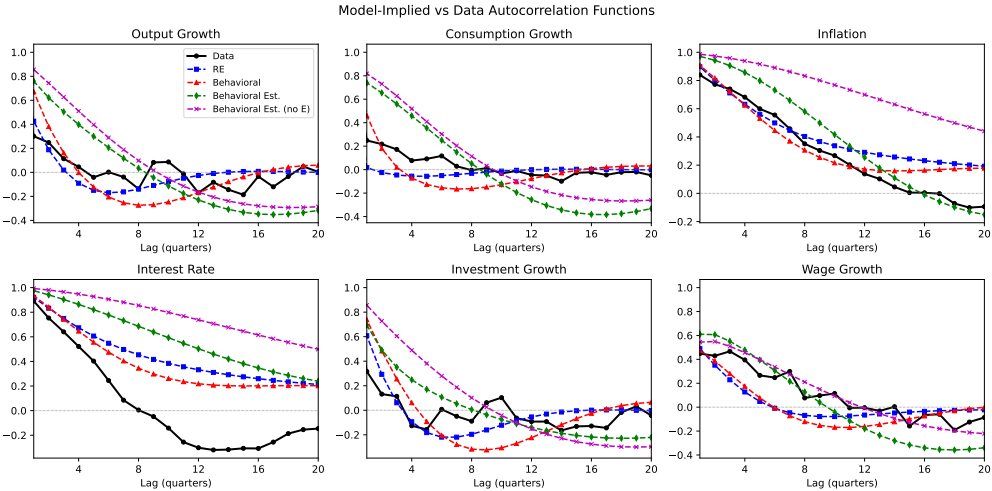
Model	Log-lik.	Behavioural Parameters			
		θ_{hh}	θ_{prof}	γ_{hh}	γ_{prof}
Rational Expectations	183.4	0	0	0	0
Behavioural (calibrated)	206.2^{***}	0.396	0.302	−0.096	−0.148
Behavioural Est. (with surveys)	266.7^{***}	0.926	0.969	≈0	≈0
Behavioural Est. (no surveys)	267.3^{***}	0.892	0.324	0.018	−0.052

- Behavioural models all fit significantly better than the RE benchmark (*** $p < 0.01$).
- Estimated behavioural parameters suggest extreme stickiness for both households and firms, with little evidence of extrapolation — tension with micro evidence.
- No survey estimation more in line with micro evidence.

Results: Forecasting Performance



Results: Model-Implied Autocorrelations



▸ Survey expectation ACFs

Conclusion

Contribution: A parsimonious, flexible framework for departing from FIRE in sequence-space DSGE/HANK models.

Theory:

- Two parameters (θ, γ) capture both underreaction to news and overreaction to current conditions
- Nests FIRE, sticky information, noisy information, and extrapolation as special cases
- Drop-in replacement for FIRE Jacobians via $\hat{J} = f(\Lambda_h, J)$
- Maintains internal rationality

Empirical:

- Departures from FIRE improve fit of the two-asset HANK model
- Generate peristence and amplification in line with macro evidence
- But some macro-micro tensions: model and survey data favour extreme stickiness for both households and firms, and don't support much extraploation.

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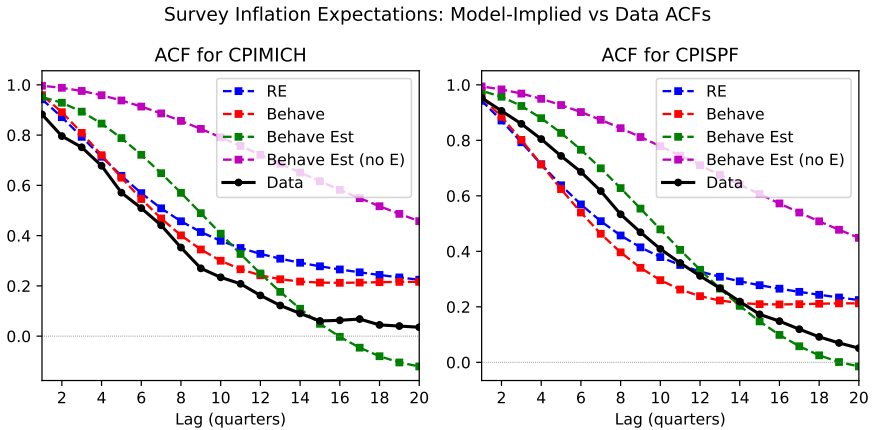
Appendix: Survey Expectation ACFs



- Model-implied vs data autocorrelation functions for Michigan Survey (households) and SPF (firms) inflation expectations
- Estimated models with survey data better match the persistence of professional forecaster expectations

Back to main ACFs

Appendix: Model Calibration

Panel A: Calibrated Parameters

Parameter		Value	Parameter		Value
ν	EIS	1	α	Capital share	0.24
ς	Frisch	0.5	δ_K	Depreciation (p.a.)	0.053
r	Real interest rate (p.a.)	0.050	μ_p	Price markup	1.06
K/Y	Capital to GDP (p.a.)	2.23	L/Y	Liquid assets to GDP (p.a.)	0.23
ξ	Intermediation spread (p.a.)	0.065	G/Y	Spending-to-GDP	0.16
qB/Y	Gov. bonds to GDP (p.a.)	0.42	$\frac{1+r}{1+r-\delta}$	Debt maturity (yrs)	5
ψ	Tax response to debt (p.a.)	0.1	ϕ_π	Taylor rule coefficient	1.5

Panel B: Household Heterogeneity

Group g	1	2	3	4	5	6
Description	Bot. 50%	Next 20%	Next 10%	Next 10%	Next 5%	Top 5%
Population share (μ_g)	0.500	0.200	0.100	0.100	0.050	0.050
Illiquid asset share	2.7%	7.0%	7.0%	13.0%	12.2%	58.0%
Labor income share	26.7%	18.3%	10.8%	14.4%	11.0%	18.8%
Discount factor β_g (p.a.)	0.905	0.919	0.933	0.946	0.950	0.975

Replicates the calibration strategy of Auclert et al. (2020), targeting aggregate US wealth, returns, bond-maturity structures, and micro iMPCs.